

How Open Source Technologies ACCELERATE INNOVATION

Open source hardware and software are the new versions of a phenomenon that has been with us since the first wandering bards added their own words to the melody of a well-loved folk tune.

Open source tools feed the dreams of amateurs and entrepreneurs alike from a deep well of well-documented designs, which are constantly being refined by the communities that use them. A growing number of traditional businesses and corporations understand that embracing them as an integral part of their business strategy may be their best hope for surviving the disruption and disintermediation that open source technologies often bring to tightly-held markets



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At first glance, the idea of freely-shared intellectual property (IP) would appear to contradict the fundamental principles of a competitive, market-driven economy. Open standards that enable interoperability between competing products are acknowledged as tools for building larger markets, but tightly-held IP and proprietary knowledge have been considered the key ingredients for innovation and success in technology-based markets from the steam age onwards.

This has begun to change over the past two or three decades, as the growing complexity of many technologies has given rise to several trends that threaten to slow the pace of innovation and market evolution. Some of these trends are:

- Many products are built upon layers of basic hardware and software functionalities that are so complex it would be impractical or impossible for most companies to develop these themselves.

- Similarly, resources needed to detect and fix the bugs in those technologies, or to update them to meet evolving market requirements, are beyond the resources of all but the largest single company.

- Innovation is further discouraged by corporations that routinely create or acquire IP with the sole purpose of creating barriers to market entry for potential competitors.

These issues have helped drive up the cost of creating or licensing many essential IP elements to the point where these make many ventures less lucrative, or even impractical. Paradoxically, the stifling business conditions these created also served as the stimulus for the emergence and adoption of open, royalty-free technologies that are developed and are supported by the people who use them.

This emergence occurred in three distinct phases, beginning with open source software. The open hardware movement that followed somewhat later adapted